

Regression 2

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Topics for this lecture

- Predictions (with uncertainty)
- Significance of regression, the F-test
- Interaction effects
- Model checking
- Polynomial regression
- Transformations of the response
- Variable selection

The body data

Let's again look at the body data and the model:

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i} + \epsilon_i$$

with the usual assumption of independent errors distributed as $\epsilon_i \sim N(0, \sigma^2)$. Here y_i is the body weight of person i , x_{1i} the height, x_{2i} is the circumference and x_{3i} is the age of the same person.

Our model with 3 predictors

```
##  
## Call:  
## lm(formula = Weight ~ Height + Circumference + Age, data = bodydata,  
##      subset = 1:20)  
##  
## Residuals:  
##      Min      1Q  Median      3Q      Max   
## -5.4088 -3.3900 -0.9123  3.0403  6.9827   
##  
## Coefficients:  
##              Estimate Std. Error t value Pr(>|t|)      
## (Intercept)  -66.5785    30.0098  -2.219  0.041330 *     
## Height         0.4768     0.1600   2.981  0.008831 **    
## Circumference  0.7769     0.1858   4.181  0.000706 ***   
## Age           -0.2094     0.3731  -0.561  0.582352      
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
## Residual standard error: 4.259 on 16 degrees of freedom  
## Multiple R-squared:  0.6852, Adjusted R-squared:  0.6262   
## F-statistic: 11.61 on 3 and 16 DF,  p-value: 0.0002729
```

Prediction

A fitted model can be used for prediction of new observations

According to model 4, what is the predicted Weight of some person with these values?

```
## Height Circumference Age
## 1 171 76 30
```

$$\hat{y} = -66.579 + 0.4768 \cdot 171 + 0.7769 \cdot 76 - 0.2094 \cdot 30$$
$$= 67.72$$

```
newobs <- data.frame('Height'=171, 'Circumference'=76, 'Age'=30)
predict.lm(LinearModel.4, newdata=newobs)
```

```
## 1
## 67.7196
```

Confidence in predictions

Confidence interval for

$E(Y|x_1, \dots, x_p) \in \widehat{\mu}(\mathbf{x}_{\text{new}}) \pm t_{\alpha/2, n-p-1} \cdot \text{SE}(\widehat{\mu}(\mathbf{x}_{\text{new}}))$ where $\text{SE}(\widehat{\mu}(\mathbf{x}_{\text{new}}))$ accounts for uncertainty in estimation.

```
predict.lm(LinearModel.4, newobs, level=0.95, interval=c("confidence"))
```

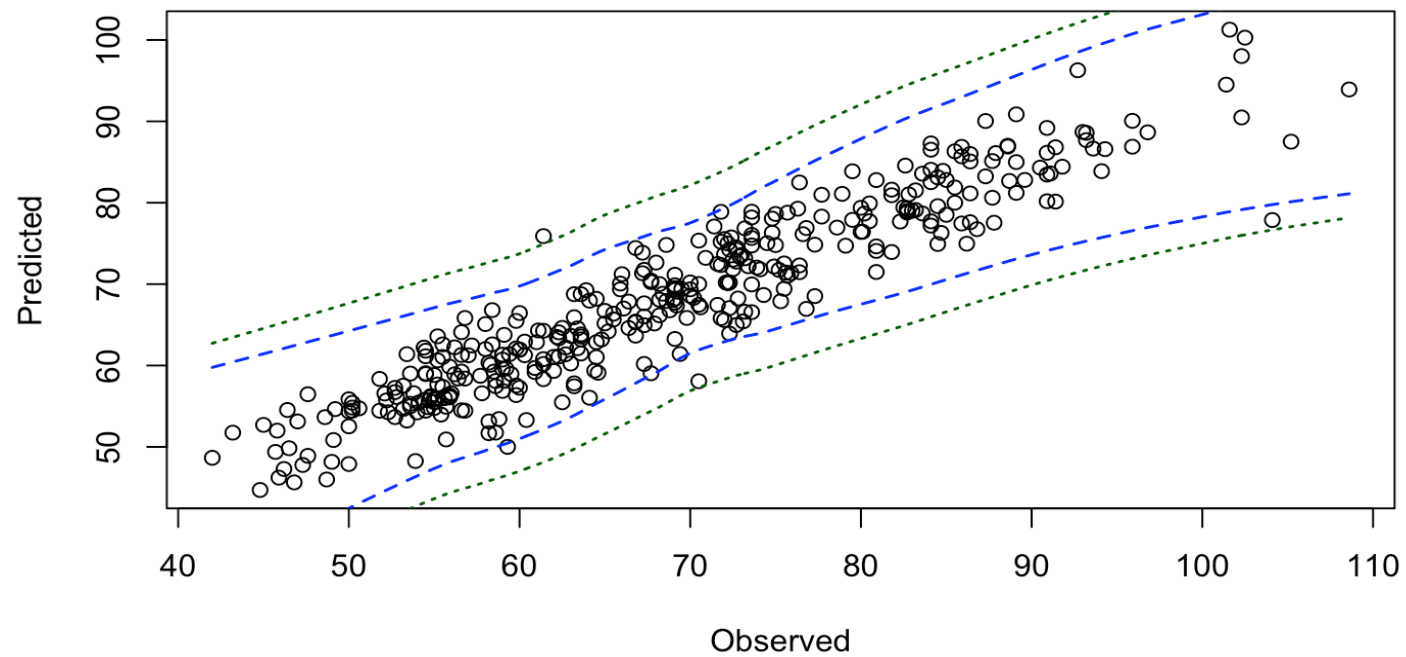
```
##          fit      lwr      upr
## 1 67.7196 60.98252 74.45668
```

Prediction interval for $Y|x_1, \dots, x_p \in \widehat{\mu}(\mathbf{x}_{\text{new}}) \pm t_{\alpha/2, n-p-1} \text{SE}_{\text{pred}}$

```
predict.lm(LinearModel.4, newobs, level=0.95, interval=c("prediction"))
```

```
##          fit      lwr      upr
## 1 67.7196 56.45463 78.98456
```

Predicted vs observed - plot



Check predictions' errors

```
##           fit      lwr      upr
## 25 73.94436 64.06824 83.82047
## 26 82.80148 72.72603 92.87693
## 27 81.03059 71.44393 90.61725
## 28 71.46852 61.81893 81.11811
## 29 65.93065 55.27452 76.58677
## 31 76.31124 66.90951 85.71296
## 32 69.35871 59.61763 79.09979
## 33 75.73802 65.67258 85.80346
## 34 82.52507 72.84459 92.20555
## 35 67.71321 57.86212 77.56431
## 37 73.86325 64.29154 83.43496
## 39 74.81258 65.46899 84.15617
```

```
## [1] " RMSE   observed: 3.8092"
```

```
## [1] " RMSE   predictions: 4.8026"
```


Checking the prediction intervals

How many observations fell within the prediction intervals?

```
largerthanlower <- (newWeight > pred[,2])  
smallerthanupper <- (newWeight < pred[,3])  
inInterval <- which(largerthanlower & smallerthanupper)  
print(length(inInterval))
```

```
## [1] 381
```

Proportion of observations within prediction intervals:

```
length(inInterval)/length(newWeight)
```

```
## [1] 0.9844961
```

What should we expect?

SSE, SST, SSR again

- Total Sum of Squares: $SST = \sum_{i=1}^n (y_i - \bar{y})^2$
- Error Sum of Squares: $SSE = \sum_{i=1}^n (y_i - \widehat{\mu}(x_i))^2$
- Regression Sum of Squares: $SSR = \sum_{i=1}^n (\widehat{\mu}(x_i) - \bar{y})^2$

It can be shown:

$$SST = SSE + SSR \quad \Rightarrow \quad R^2 = 1 - \frac{SSE}{SST} = \frac{SSR}{SST}$$

Significance of regression?

We have seen how we can test the significance of single predictor terms with T-tests in the previous lesson.

We may also test the overall significance of the regression model.

Recall from earlier that $R^2 = 1 - \frac{SSE}{SST} = \frac{SSR}{SST}$ is a measure of model fit. If the model can explain much of the observed variability in y , then SSR is close to SST and big compared to SSE.

Can be shown that a convenient test observator, typically named F (for Fisher) is:

$$F = \frac{SSR/p}{SSE/(n - p - 1)}$$

The F-test for overall significance of regression

Hypothesis:

$H_0 : \beta_1 = \beta_2 = \cdots = \beta_p = 0$ vs $H_1 : \text{at least one } \beta_j \neq 0$.

Reject the null hypothesis at test level α if F is large, and more specifically if

$$F > F_{\alpha, p, n-p-1}$$

Let's see how this turns out for the body data model.

The F-test illustration

Recall

```
##  
## Call:  
## lm(formula = Weight ~ Height + Circumference + Age, data = bodydata,  
##      subset = 1:20)  
##  
## Residuals:  
##      Min      1Q  Median      3Q      Max   
## -5.4088 -3.3900 -0.9123  3.0403  6.9827   
##  
## Coefficients:  
##              Estimate Std. Error t value Pr(>|t|)      
## (Intercept)  -66.5785    30.0098  -2.219  0.041330 *     
## Height         0.4768     0.1600   2.981  0.008831 **    
## Circumference  0.7769     0.1858   4.181  0.000706 ***   
## Age          -0.2094     0.3731  -0.561  0.582352      
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
## Residual standard error: 4.259 on 16 degrees of freedom  
## Multiple R-squared:  0.6852, Adjusted R-squared:  0.6262   
## F-statistic: 11.61 on 3 and 16 DF,  p-value: 0.0002729
```

The body data

```
body1 <- lm(Weight ~ Height + Circumference + Age, data=bodydata, subset = 1:20)
Atab <- anova(body1)
```

```
## Analysis of Variance Table
```

```
##
```

```
## Response: Weight
```

```
##           Df Sum Sq Mean Sq F value    Pr(>F)
## Height      1 308.31   308.31  16.9984 0.0007973 ***
## Circumference 1 317.76   317.76  17.5190 0.0006993 ***
## Age         1   5.72    5.72   0.3151 0.5823519
## Residuals   16 290.20    18.14
```

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
## SST = 921.988 , SSR = 631.7836 ;
```

```
## Cumulative SST 308.3131 626.0686 631.7836 921.988 ;
```

```
## F = 11.61083
```

The body data - Compare with previous

```
body1 <- lm(Weight ~ Circumference + Height + Age, data=bodydata, subset = 1:20)
Atab <- anova(body1)
```

```
## Analysis of Variance Table
```

```
##
```

```
## Response: Weight
```

```
##           Df Sum Sq Mean Sq F value    Pr(>F)
## Circumference  1  467.18   467.18  25.7575 0.0001125 ***
## Height         1  158.89   158.89   8.7599 0.0092221 **
## Age            1    5.72     5.72   0.3151 0.5823519
## Residuals      16 290.20     18.14
```

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
## SST = 921.988 , SSR = 631.7836 , F = 11.61083
```


Interaction effects

Interaction effects occur when the effect of one variable on the response variable depends on the level/value of another variable.

Consider the model:

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i} + \beta_4 x_{2i} x_{3i} + \epsilon_i$$

here we have expanded the model with an interaction term between x_2 and x_3 (between circumference and age). How should we interpret this model?

How to fit the model in RStudio

Now fit on all the data we have

```
data(bodydata)
body2 <- lm(Weight ~ Height + Circumference + Age +
            Circumference:Age, data=bodydata)
summary(body2)
```

The output with all data

```
##
## Call:
## lm(formula = Weight ~ Height + Circumference + Age + Circumference:Age,
##     data = bodydata)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -15.1735  -2.5820  -0.1732   2.6033  22.5148
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -93.579875    6.030390  -15.518  < 2e-16 ***
## Height         0.414057    0.027243   15.199  < 2e-16 ***
## Circumference  1.244437    0.070990   17.530  < 2e-16 ***
## Age            0.611383    0.159085    3.843 0.000141 ***
## Circumference:Age -0.009476  0.001990   -4.763 2.67e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.252 on 402 degrees of freedom
## Multiple R-squared:  0.8989, Adjusted R-squared:  0.8979
## F-statistic: 893.8 on 4 and 402 DF,  p-value: < 2.2e-16
```

Model checking

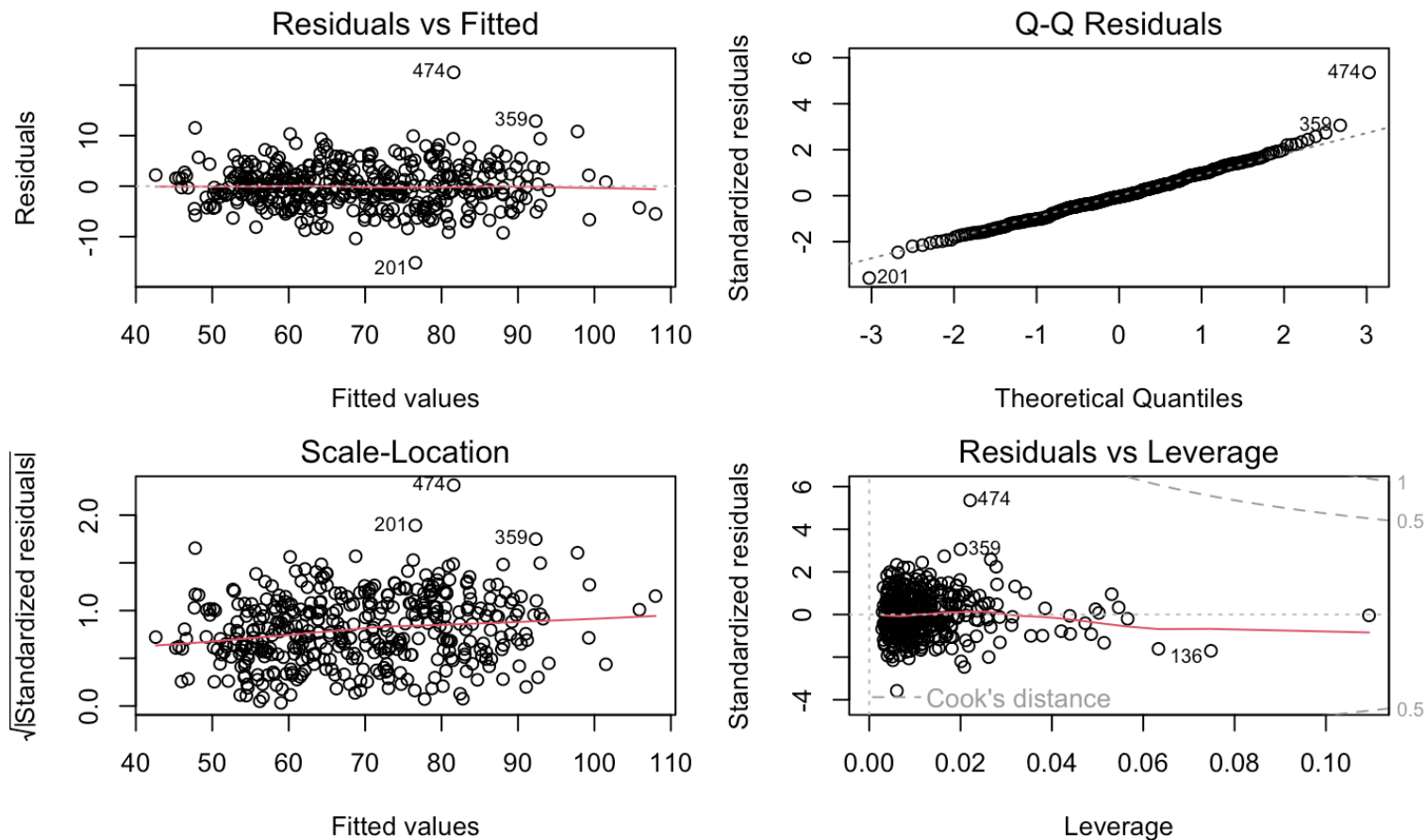
In standard linear regression we typically assume:

- Linearity
- Constant variance
- Independent observations
- Normality

We should check that these assumptions are OK. Typically we do this by examining the residuals

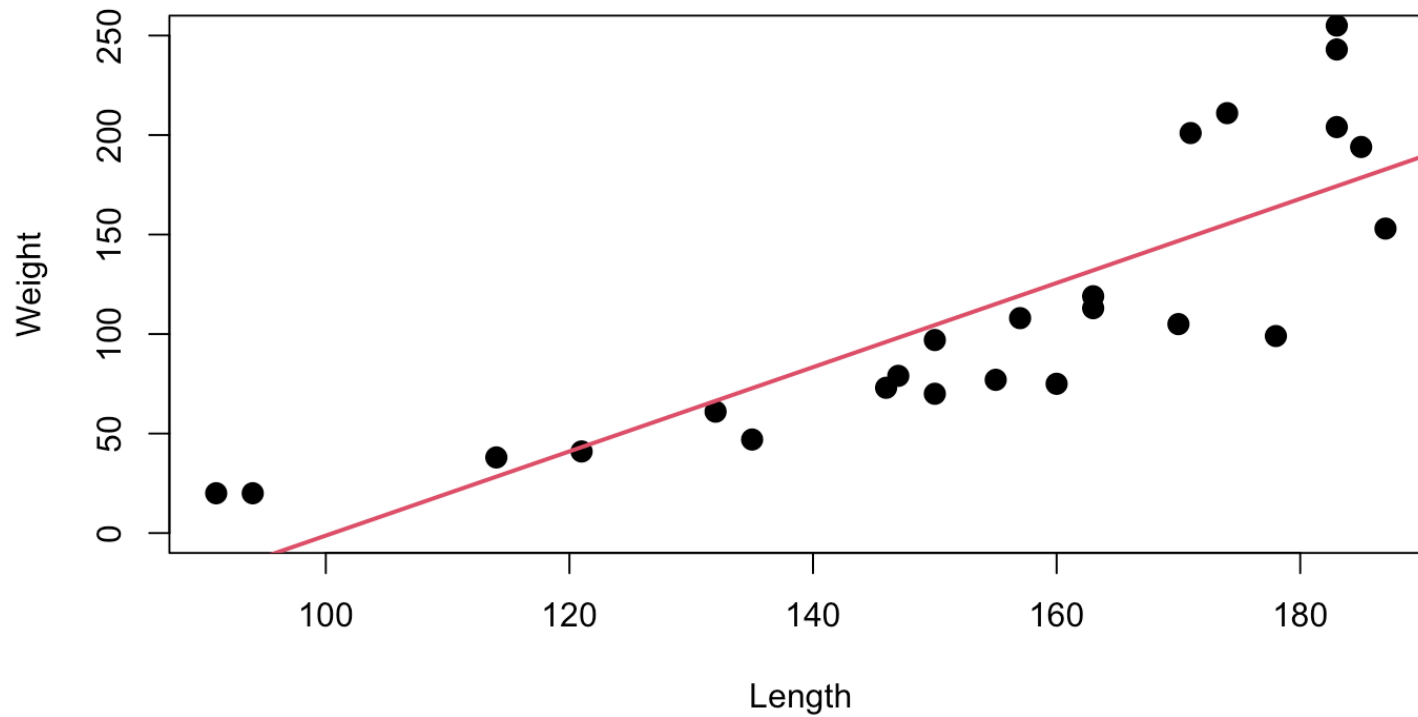
Residual plots

```
par(mfrow=c(2,2), mar=c(4.5,4,1.5,1.5))  
plot(body2, ask=FALSE)
```



Draw Q-Q plot and std residuals

Another example - Bears

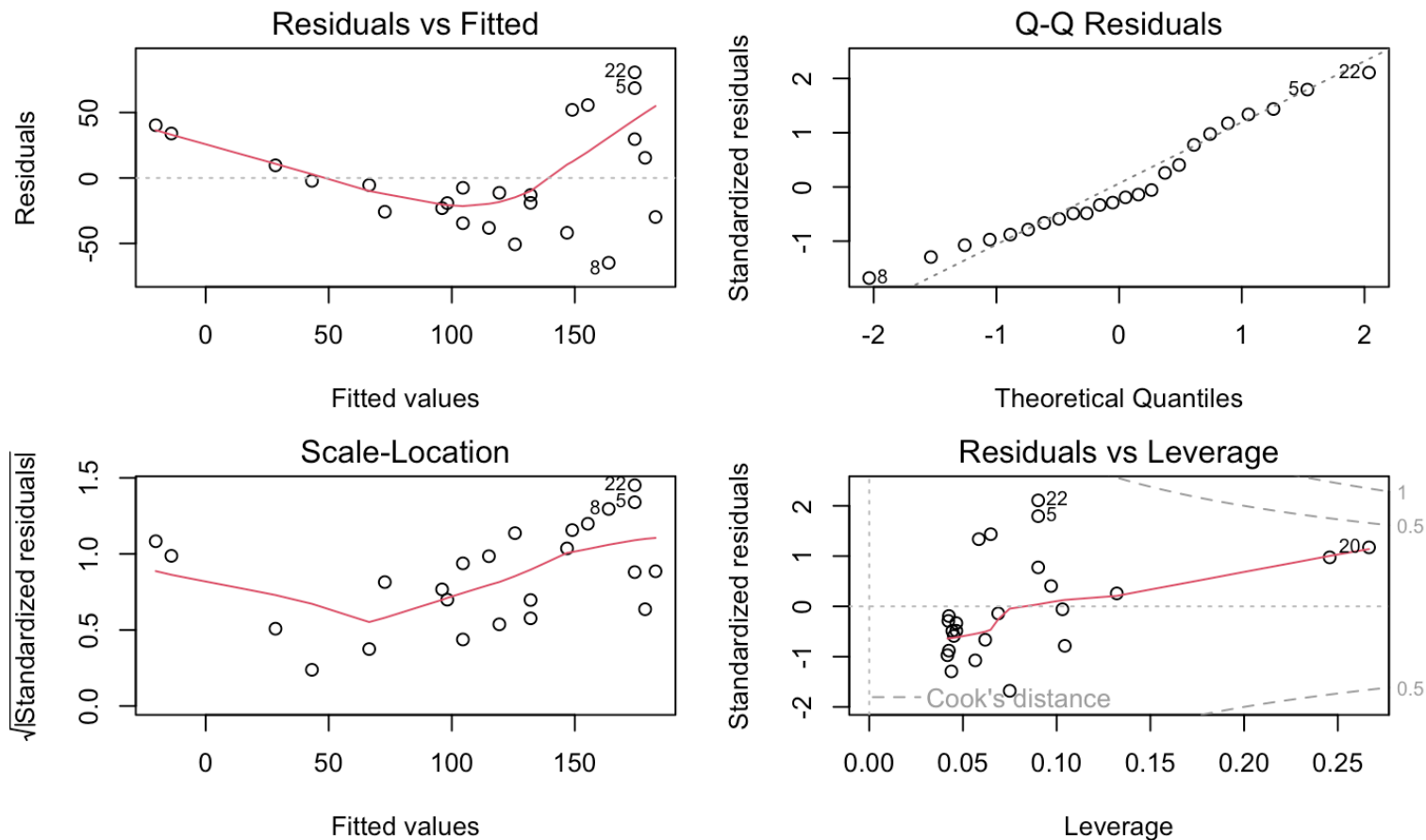


Summary

```
##  
## Call:  
## lm(formula = Weight ~ Length, data = bears)  
##  
## Residuals:  
##      Min       1Q   Median       3Q      Max   
## -64.76 -26.78  -9.42   30.74   80.67   
##  
## Coefficients:  
##              Estimate Std. Error t value Pr(>|t|)      
## (Intercept) -212.8522    47.2818  -4.502 0.000177 ***  
## Length       2.1158     0.3027   6.989 5.15e-07 ***  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
## Residual standard error: 40.1 on 22 degrees of freedom  
## Multiple R-squared:  0.6895, Adjusted R-squared:  0.6754   
## F-statistic: 48.85 on 1 and 22 DF,  p-value: 5.147e-07
```


Residual plots

```
par(mfrow=c(2,2), mar=c(4.5,4,1.5,1.5))  
plot(bears1, ask=FALSE)
```



Outliers and influence statistics

- Leverage
- Standardized residuals
- Deleted residuals (PRESS residuals) (predict the deleted leave out one point)
- Cook's distance (combines leverage and standardized residuals) > 1 , can be dangerous

Explain leverage

Polynomial regression

Non-linear relationships may be modelled by polynomial regression.

Bear example

- y = Weight
- x_1 = Length
- x_1^2 = Length²

A multiple LINEAR model (linear in the β 's)

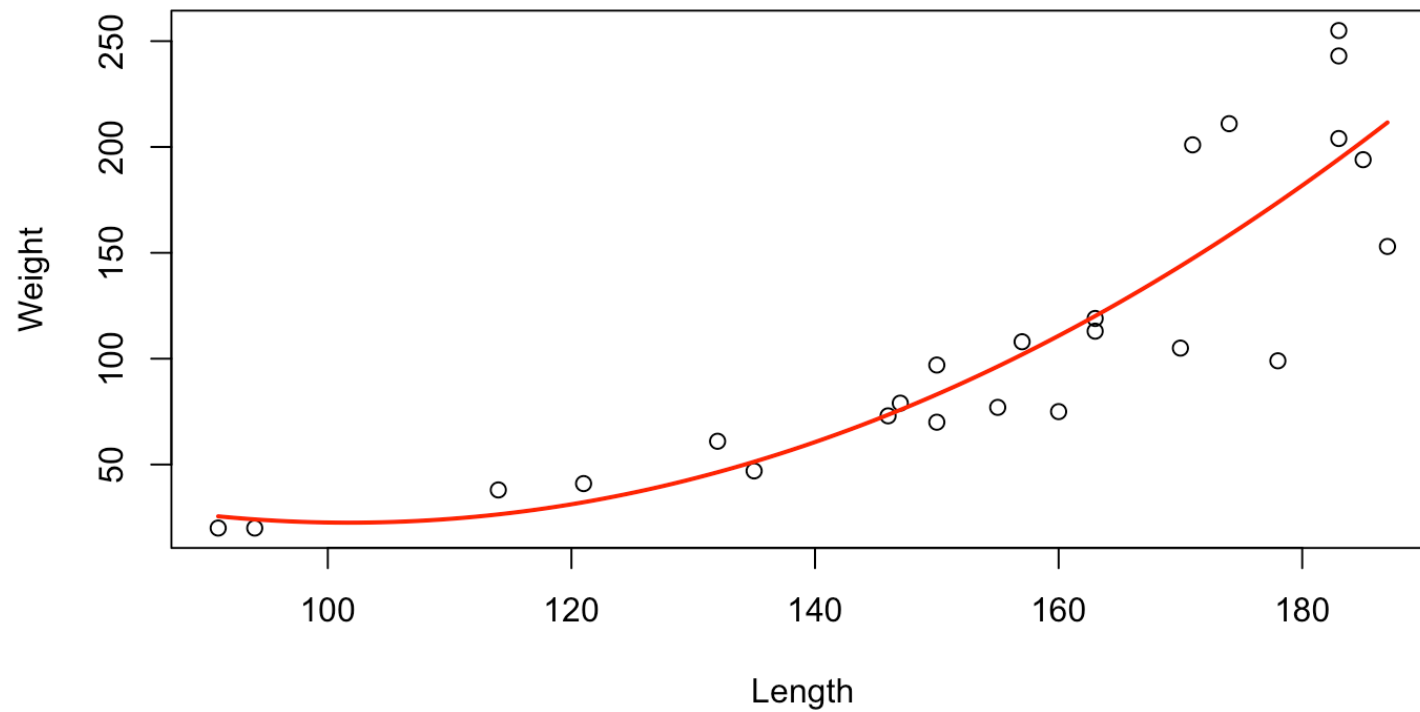
$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_1^2 + \epsilon$$

Fitting the model in R

```
bears2 <- lm(Weight ~ Length + I(Length^2), data=bears)
summary(bears2)
```

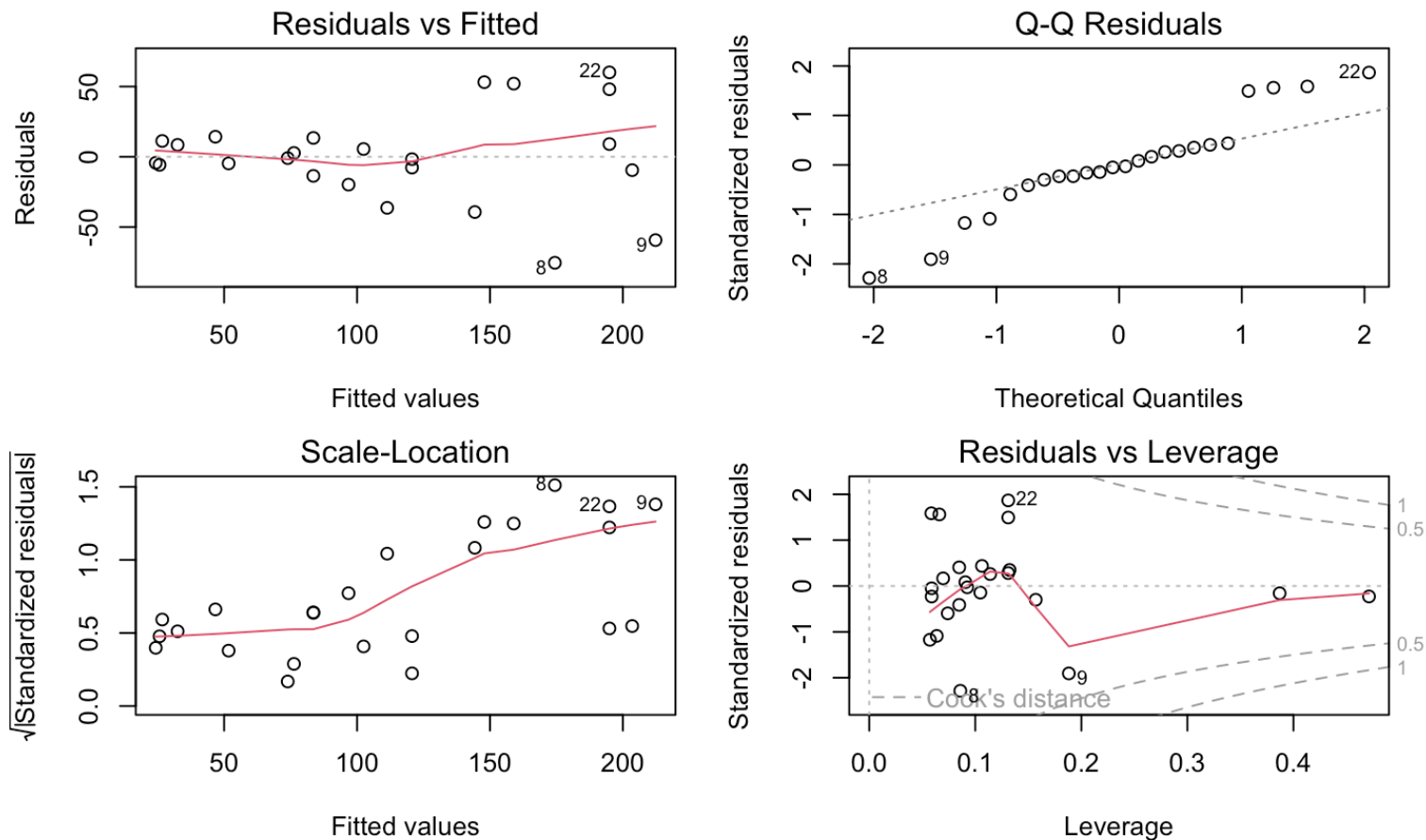
```
##
## Call:
## lm(formula = Weight ~ Length + I(Length^2), data = bears)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -75.411 -10.514  -1.302   11.809   60.075
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  291.623760  175.892266   1.658  0.11219
## Length       -5.289645   2.525406  -2.095  0.04852 *
## I(Length^2)   0.026018   0.008825   2.948  0.00768 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 34.52 on 21 degrees of freedom
## Multiple R-squared:  0.7804, Adjusted R-squared:  0.7595
## F-statistic: 37.31 on 2 and 21 DF, p-value: 1.223e-07
```

A fitted line plot

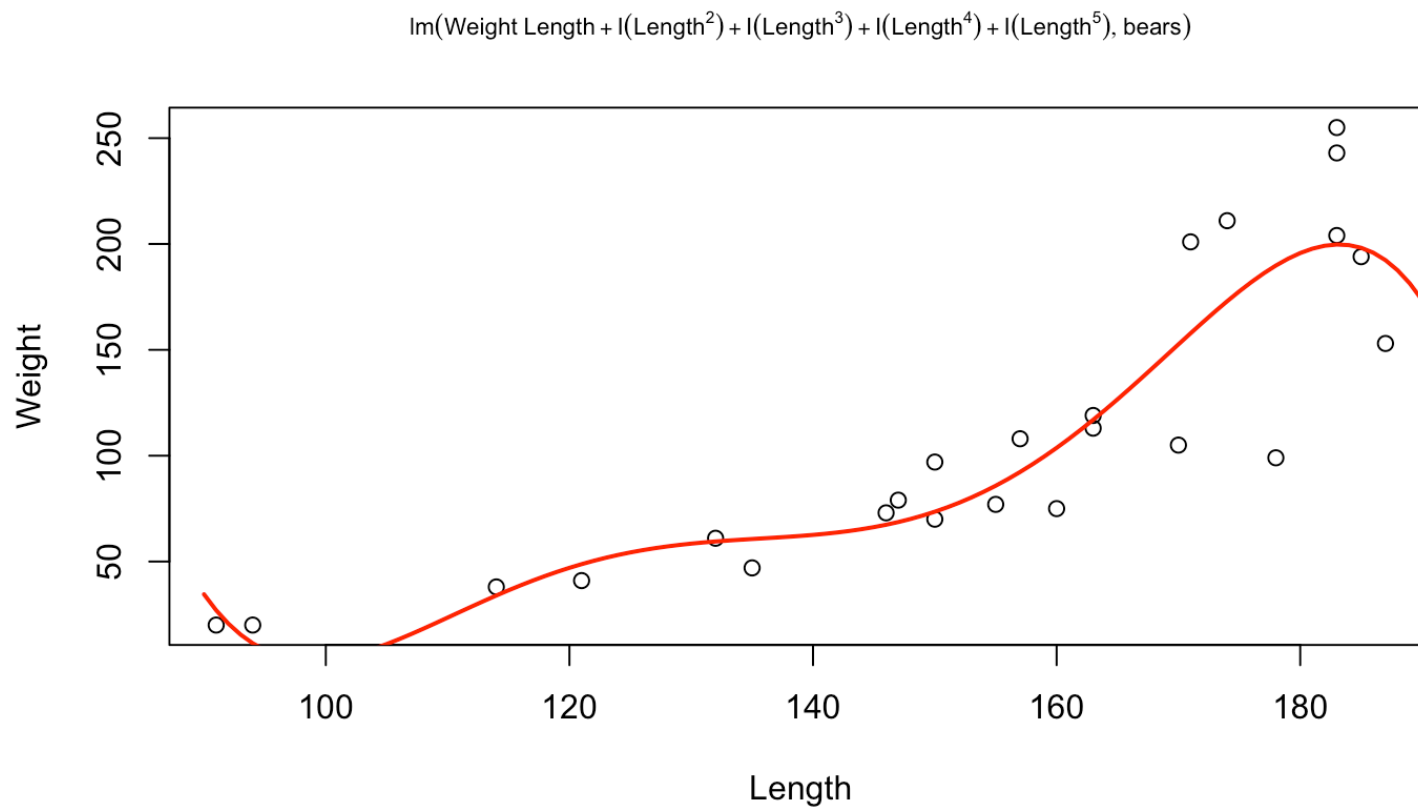


Model checking

```
par(mfrow=c(2,2), mar=c(4.5,4,1.5,1.5))  
plot(bears2, ask=FALSE)
```

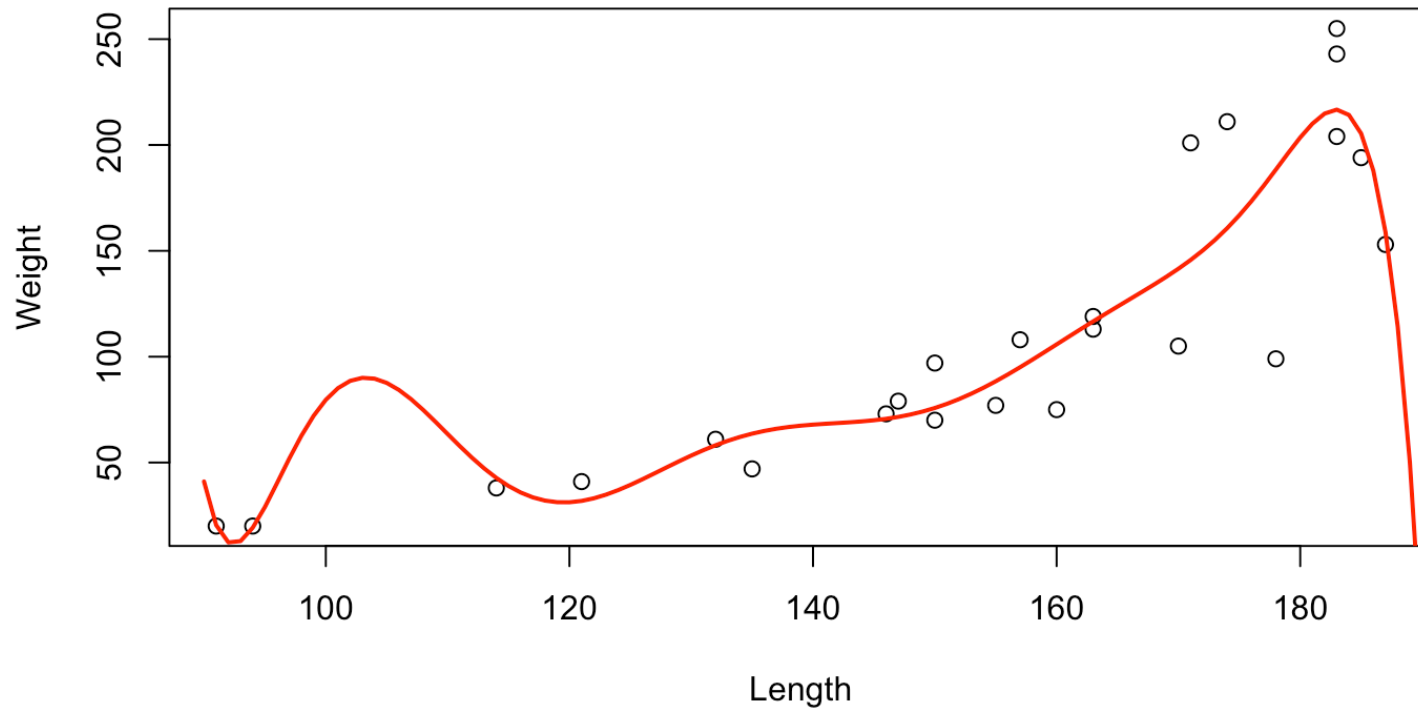


Careful! Over-fitting!

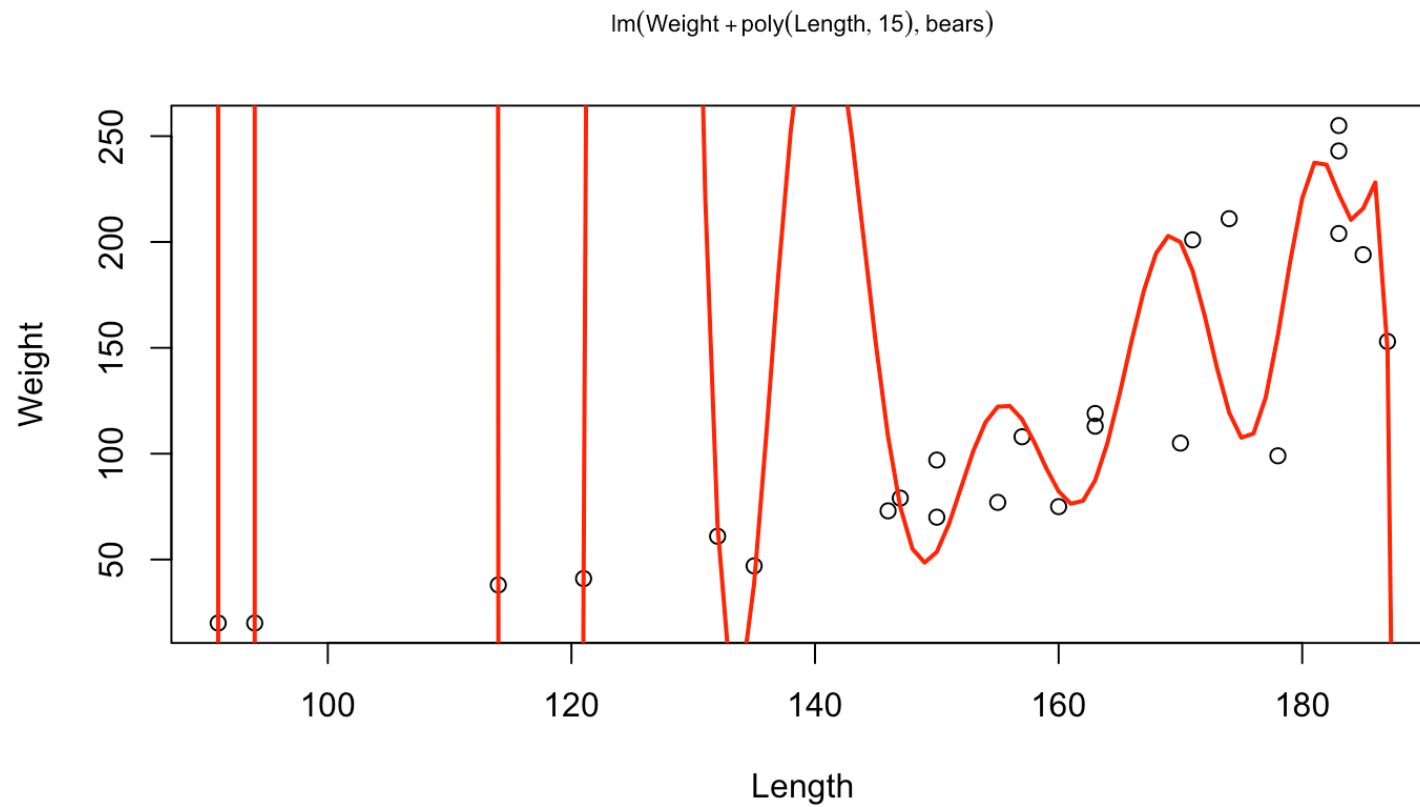


Careful! Over-fitting!

$\text{lm}(\text{Weight} \sim \text{Length} + \text{I}(\text{Length}^2) + \text{I}(\text{Length}^3) + \text{I}(\text{Length}^4) + \text{I}(\text{Length}^5) + \text{I}(\text{Length}^6) + \text{I}(\text{Length}^7) + \text{I}(\text{Length}^8) + \text{I}(\text{Length}^9) + \text{I}(\text{Length}^{10}), \text{bears})$



Careful! Over-fitting!



Transformations

Deviations from model assumptions may be fixed by transforming y and/or x .

Some transformations which may improve linearity and stabilize variance are

- $\tilde{y} = \log(y)$
- $\tilde{x} = \log(x)$
- $\tilde{y} = \sqrt{y}$

Then use $\tilde{y}$ and/or $\tilde{x}$ instead of y and x to fit the model

The log(Weight) response model

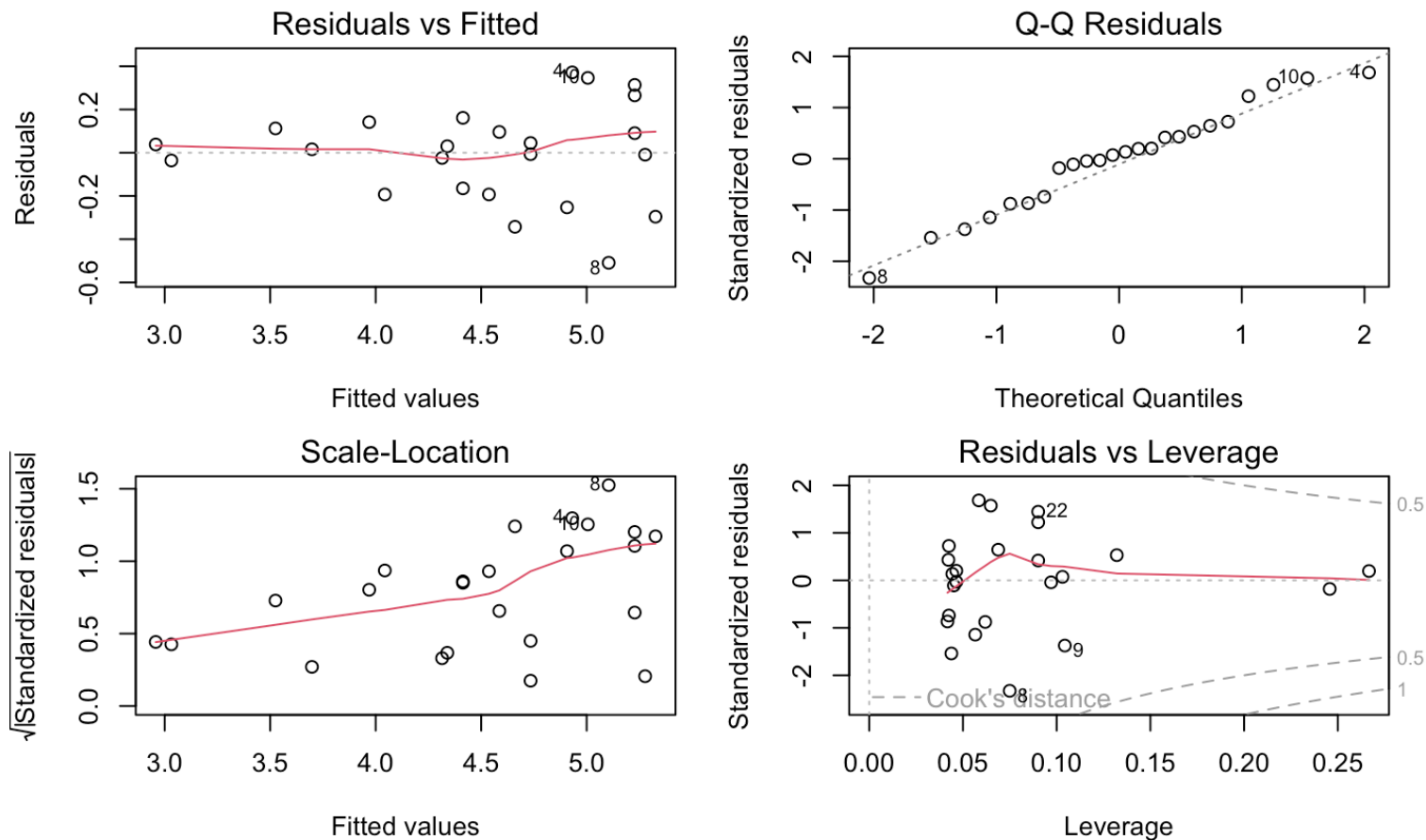
```
bears4 <- lm(log(Weight) ~ Length, data=bears)
summary(bears4)
```

```
##
## Call:
## lm(formula = log(Weight) ~ Length, data = bears)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.50918 -0.17192  0.02291  0.11978  0.37173
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.712120   0.268342   2.654   0.0145 *
## Length       0.024675   0.001718  14.363 1.17e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2276 on 22 degrees of freedom
## Multiple R-squared:  0.9036, Adjusted R-squared:  0.8992
## F-statistic: 206.3 on 1 and 22 DF,  p-value: 1.173e-12
```

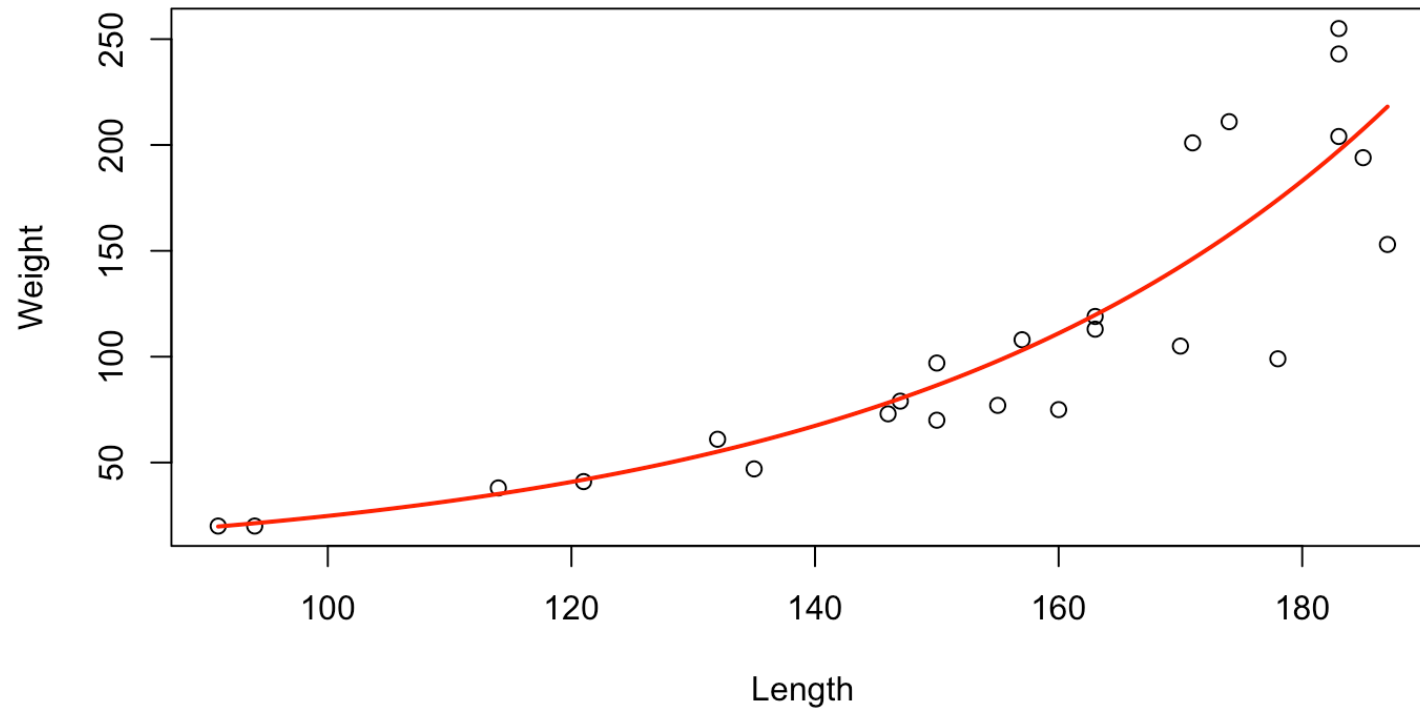
Explain the transformation and back-transformation

Residual plots

```
par(mfrow=c(2,2), mar=c(4.5,4,1.5,1.5))  
plot(bears4, ask=FALSE)
```



A fitted line plot



Variable selection

Variable selection is the process of selecting a subset of variables that are most relevant for the response in some way. Various approaches. Some are based on hypothesis testing, some on criteria (like Akaike's Information Criteria - AIC, Bayesian information criterion - BIC, adjusted R^2_{adj} , etc)

Sometimes combined with stepwise approaches like:

- Forward selection
- Backward selection
- Forward/Backward
- Backward/Forward
- All subsets selection

```
library(mixlm)
```

```
##
```

```
## Attaching package: 'mixlm'
```


Stepwise methods - Backward selection

```
body1 <- lm(Weight ~ Height + Circumference + Age, data=bodydata, subset=1:50)
bs <- backward(body1, alpha=0.05)
```

```
## Backward elimination, alpha-to-remove: 0.05
```

```
##
```

```
## Full model: Weight ~ Height + Circumference + Age
```

```
##
```

##	Step	RSS	AIC	R2pred	Cp	F value	Pr(>F)
## Age	1	964.52	153.98	0.71627	3.3702	1.3702	0.2478

Stepwise methods - Backward selection

```
summary(bs)
```

```
##
```

```
## Call:
```

```
## lm(formula = Weight ~ Height + Circumference, data = bodydata,  
##      subset = 1:50)
```

```
##
```

```
## Residuals:
```

```
##      Min        1Q    Median        3Q        Max  
## -9.6399 -2.5548  0.2676   3.1122   9.9851
```

```
##
```

```
## Coefficients:
```

```
##              Estimate Std. Error t value Pr(>|t|)  
## (Intercept)  -88.77490   16.95517  -5.236 3.77e-06 ***  
## Height         0.43630    0.09669   4.512 4.28e-05 ***  
## Circumference  1.09158    0.11835   9.223 4.07e-12 ***
```

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
```

```
## s: 4.53 on 47 degrees of freedom
```

```
## Multiple R-squared:  0.7534,
```

```
## Adjusted R-squared:  0.7429
```

Stepwise methods - Forward selection

```
body1 <- lm(Weight ~ Height + Circumference + Age, data=bodydata, subset=1:50)
fs <- forward(body1, alpha=0.05)
```

```
## Forward selection, alpha-to-enter: 0.05
```

```
##
```

```
## Full model: Weight ~ Height + Circumference + Age
```

```
##
```

```
##           Step      RSS      AIC  R2pred      Cp F value      Pr(>F)
## Circumference    1 1382.37 169.98 0.60830 21.8918  87.788 2.043e-12 ***
## Height           2   964.52 153.98 0.71627  3.3702  20.361 4.281e-05 ***
## ----
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Stepwise methods - Forward selection

```
summary(fs)
```

```
##  
## Call:  
## lm(formula = Weight ~ Circumference + Height, data = bodydata,  
##      subset = 1:50)  
##  
## Residuals:  
##      Min        1Q    Median        3Q        Max   
## -9.6399 -2.5548  0.2676   3.1122   9.9851   
##  
## Coefficients:  
##              Estimate Std. Error t value Pr(>|t|)      
## (Intercept)  -88.77490    16.95517   -5.236 3.77e-06 ***  
## Circumference   1.09158     0.11835    9.223 4.07e-12 ***  
## Height         0.43630     0.09669    4.512 4.28e-05 ***  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
## s: 4.53 on 47 degrees of freedom  
## Multiple R-squared:  0.7534,  
## Adjusted R-squared:  0.7429
```

Best subsets method

If the number of candidate variables, p , is limited, e.g. ≤ 20 , we may compare all possible models

There are 2^p possible models

Compare all model fits by AIC, BIC, R_{adj}^2

Choose model based on:

- Best-fit - simplicity trade-off

Best subsets method - Example

```
body1 <- lm(Weight ~ Height + Circumference + Age, data=bodydata, subset=1:50)
bs <- best.subsets(body1)
```

##		Height	Circumference	Age	RSS	R2	R2adj	Cp
## 1	(1)			*	13731.668	0.80903505	0.80856354	317.73818
## 1	(2)	*			34547.127	0.51955652	0.51837024	1410.28546
## 1	(3)			*	69391.877	0.03497401	0.03259123	3239.19241
## 2	(1)	*		*	8285.670	0.88477201	0.88420157	33.89246
## 2	(2)			*	12297.068	0.82898589	0.82813929	244.43992
## 2	(3)	*		*	32876.376	0.54279148	0.54052808	1324.59227
## 3	(1)	*		*	7678.048	0.89322215	0.89242728	4.00000